

NTT-Friendly Pairing Curve Construction with Readability-Aware Prime Selection over a 1024-bit Field

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Abstract—Pairing-friendly elliptic curves are fundamental to advanced cryptographic protocols such as BLS signatures and zero-knowledge proof systems. However, the traditional Cocks-Pinch construction produces primes with no guaranteed algebraic structure, limiting their utility for NTT-based proof systems and hindering parameter auditability. We present three contributions: (1) a modified Cocks-Pinch algorithm that constrains the cyclotomic seed to guarantee two-adicity of at least 36 in both the base field and scalar field primes, exceeding the industry-standard BLS12-381 curve; (2) a multi-criteria readability scoring system that selects human-auditable primes from the valid candidate space without weakening security; and (3) Curve1024, a concrete 1024-bit pairing-friendly curve ($E : y^2 = x^3 + 41$, $k = 18$, 256-bit ECDLP security) with full parameter set and generator point published for public verification. A complete Rust implementation with 36 passing tests demonstrates practical signing operations and validates resistance against MOV, Anomalous, and TNFS attacks.

Index Terms—elliptic curve cryptography, pairing-friendly curves, Cocks-Pinch method, NTT-friendly primes, parameter auditability

I. INTRODUCTION

Pairing-friendly elliptic curves underpin a growing class of cryptographic protocols, from BLS aggregate signatures [1] to succinct non-interactive zero-knowledge proofs (zk-SNARKs) [2], [3]. These protocols require curves whose scalar field and base field support efficient polynomial arithmetic via the Number Theoretic Transform (NTT), which in turn demands that both field primes satisfy $q \equiv 1 \pmod{2^s}$ for a sufficiently large s .

The Cocks-Pinch method [4] offers unmatched flexibility in constructing pairing-friendly curves with arbitrary embedding degrees. Unlike parametric families such as BN [5] or BLS12 [6], the Cocks-Pinch approach can target specific security levels and field sizes. However, the traditional algorithm treats NTT-friendliness as an afterthought—primes are generated with random two-adicity, typically only 1–3 bits, rendering them unsuitable for modern proof systems.

A separate but related concern is *parameter auditability*. The Dual_EC_DRBG incident [7] demonstrated that opaque cryptographic constants can conceal backdoors. While the Cocks-Pinch method is inherently transparent (all parameters derive from a public seed), the resulting primes are typically inscrutable hexadecimal blobs that resist manual inspection.

Contributions. This paper presents three results:

1. **Improved Cocks-Pinch with NTT constraints.** We modify the Cocks-Pinch construction to *guarantee* high two-adicity in both field primes by constraining the cyclotomic seed T . Choosing $T \equiv 0 \pmod{2^{12}}$ yields $\nu_2(r-1) \geq 36$ deterministically, while an extended lift search ensures $\nu_2(p-1) \geq 36$. The resulting two-adicity of 36 exceeds BLS12-381’s value of 32.
2. **Readability-aware prime selection.** We introduce a multi-criteria scoring system that evaluates candidate primes on eight structural properties (binary sparseness, zero limbs, hex density, etc.) and selects the most human-auditable parameters without sacrificing security.
3. **Curve1024: a concrete 1024-bit pairing-friendly curve.** We instantiate the above techniques to produce a complete, publicly verifiable curve with embedding degree $k = 18$, scalar field order $r \approx 2^{512}$ (256-bit ECDLP security), and equation $E : y^2 = x^3 + 41$. The full parameter set—including generator point—is published and validated by a Rust implementation with 36 passing tests.

Organization. Section 2 reviews background on pairing-friendly curves and the CM method. Section 3 presents our NTT-friendly Cocks-Pinch modification. Section 4 describes the readability scoring system. Section 5 presents the Curve1024 parameters and evaluates the implementation. Section 6 discusses related work, and Section 7 concludes.

II. PRELIMINARIES

A. Elliptic Curves and Pairings

Let $\mathbb{F}_p$ be a prime field with $p > 3$. An elliptic curve in Short Weierstrass form is $E : y^2 = x^3 + ax + b$ with $4a^3 + 27b^2 \neq 0$. The set $E(\mathbb{F}_p)$ of rational points forms an abelian group under the chord-and-tangent law [8].

A curve is *pairing-friendly* with embedding degree k if there exists a prime-order subgroup of size r such that $r \mid p^k - 1$ but $r \nmid p^i - 1$ for $0 < i < k$. This enables a bilinear map $e : \mathbb{G}_1 \times \mathbb{G}_2 \rightarrow \mathbb{G}_T$ where $\mathbb{G}_T \subset \mathbb{F}_{p^k}^*$, which is the foundation of BLS signatures and zk-SNARKs.

B. Complex Multiplication Method

The CM method constructs curves with a prescribed group order. Given a CM discriminant $D < 0$, the existence of a curve over $\mathbb{F}_p$ with Frobenius trace t requires the Diophantine equation:

$$4p = t^2 + |D| \cdot y^2$$

to have an integer solution. The j -invariant is obtained from the Hilbert class polynomial $H_D(x)$. For $D = -3$, we have $H_{-3}(x) = x$, giving $j = 0$ and the simple curve form $y^2 = x^3 + b$.

C. Traditional Cocks-Pinch Algorithm

The Cocks-Pinch method [4] constructs pairing-friendly curves by starting from the subgroup order r rather than the field prime p . For embedding degree $k = 18$:

1. **Generate r :** Choose a parameter T and compute $r = \Phi_{18}(T) = T^6 - T^3 + 1$. Verify that r is prime.
2. **Compute square root:** Find $\beta = \sqrt{-3} \pmod{r}$ via Tonelli-Shanks.
3. **Compute CM parameters:** For each i with $\gcd(i, 18) = 1$:

$$t_0 \equiv T^i + 1 \pmod{r}, \quad y_0 \equiv \frac{t_0 - 2}{\beta} \pmod{r}$$

4. **Lift to integers:** Search over $(h_t, h_y) \in [-20, 20]^2$ for:

$$t = t_0 + h_t \cdot r, \quad y = y_0 + h_y \cdot r$$

such that $p = (t^2 + 3y^2)/4$ is a prime of the target size.

The resulting curve has $\rho = \log p / \log r \approx 2$, which is the standard tradeoff of the Cocks-Pinch method compared to parametric families ($\rho = 1$ for BN, $\rho = 1.5$ for BLS12).

D. Two-Adicity and NTT

The *two-adicity* of a prime q is $\nu_2(q-1)$, the largest power of 2 dividing $q-1$. A field $\mathbb{F}_q$ with two-adicity s contains a primitive 2^s -th root of unity, enabling NTT of length up to 2^s . Modern ZK proof systems (Groth16 [2], PLONK [3]) require NTT for polynomial multiplication over both $\mathbb{F}_r$ (constraint evaluation) and $\mathbb{F}_p$ (pairing computation). BLS12-381 achieves two-adicity 32 in both fields [6].

III. IMPROVED COCKS-PINCH WITH NTT CONSTRAINTS

We modify the traditional Cocks-Pinch algorithm to guarantee high two-adicity in both output primes. The key insight is that the algebraic structure of the cyclotomic polynomial Φ_{18} allows deterministic control over the two-adicity of r , while an extended lift search achieves the same for p .

A. NTT-Friendly Scalar Field via Seed Constraint

Since $r = \Phi_{18}(T) = T^6 - T^3 + 1$, we observe:

$$r - 1 = T^6 - T^3 = T^3(T^3 - 1)$$

If $T \equiv 0 \pmod{2^m}$, then $T^3 \equiv 0 \pmod{2^{3m}}$. Since $T^3 - 1$ is odd (as T^3 is even), we obtain:

$$\nu_2(r - 1) = \nu_2(T^3) = 3m$$

Consequence: To guarantee $\nu_2(r - 1) \geq s$, it suffices to choose $T \equiv 0 \pmod{2^{\lceil s/3 \rceil}}$. This is a *deterministic* guarantee requiring no rejection sampling.

For our target of $s = 36$, we set $m = \lceil 36/3 \rceil = 12$, constraining $T \equiv 0 \pmod{2^{12}}$. The search space remains $2^{86-12} = 2^{74}$ candidates—more than sufficient for finding prime r .

B. NTT-Friendly Base Field via Extended Lift Search

After obtaining a valid r with the desired two-adicity, the base field prime is:

$$p = \frac{(t_0 + h_t r)^2 + 3(y_0 + h_y r)^2}{4}$$

The two-adicity of $p - 1$ depends on the lift parameters (h_t, h_y) . To ensure $\nu_2(p - 1) \geq s_p$, we extend the search grid beyond the default $[-20, 20]^2$:

$$\text{half_range} = 20 + 2^{\max(0, s_p - 3m)}$$

This extension guarantees sufficient coverage of bit patterns in $p - 1$ at positions above $3m$. For $s_p = 36$ and $m = 12$ (so $3m = 36$), the grid remains at $[-21, 21]^2$ —only marginally larger than the default.

C. Algorithm

The complete NTT-friendly Cocks-Pinch construction proceeds as follows:

Input: Embedding degree $k = 18$, target sizes $(|r|, |p|)$, minimum two-adicity s

Output: Primes (p, r) with $\nu_2(r - 1) \geq s$ and $\nu_2(p - 1) \geq s$

1. Set $m \leftarrow \lceil s/3 \rceil$ and step $\leftarrow 2^m$
2. Repeat:
 - a. Choose T as a random multiple of step in the target range
 - b. Compute $r \leftarrow T^6 - T^3 + 1$
 - c. If r is not prime or $|r| \neq \text{target}$, continue
 - d. Compute $\beta \leftarrow \sqrt{-3} \pmod{r}$ (Tonelli-Shanks)
 - e. For each i with $\gcd(i, 18) = 1$:
 - $t_0 \leftarrow T^i + 1 \pmod{r}$
 - $y_0 \leftarrow (t_0 - 2) \cdot \beta^{-1} \pmod{r}$
 - $\text{range} \leftarrow 20 + 2^{\max(0, s - 3m)}$
 - For $(h_t, h_y) \in [-\text{range}, \text{range}]^2$:
 - $t \leftarrow t_0 + h_t \cdot r$; $y \leftarrow y_0 + h_y \cdot r$
 - $p \leftarrow (t^2 + 3y^2)/4$
 - If $|p| = \text{target}$ AND $\nu_2(p - 1) \geq s$ AND p is prime: return (p, r, t, y)
3. Until max attempts reached

TABLE I
TRADITIONAL VS. IMPROVED COCKS-PINCH

Metric	Traditional	Improved
Constraint on T	None	$T \equiv 0 \pmod{2^{12}}$
$\nu_2(r-1)$	Random (1–3)	≥ 36 (guaranteed)
$\nu_2(p-1)$	Random (1–3)	≥ 36 (with ext. grid)
Grid size	$[-20, 20]^2$	$[-21, 21]^2$
Avg. attempts	100–500	3,000–6,000
Generation time	2–10 s	30–60 s
NTT support	No	Up to 2^{36}

D. Complexity Analysis

The following table compares the traditional and improved algorithms. The NTT constraint increases the average number of attempts by approximately $10\text{--}60\times$ (from $\sim 100\text{--}500$ to $\sim 3,000\text{--}6,000$), but the absolute generation time remains under one minute—a one-time cost amortized over the lifetime of the curve parameters.

E. Comparison with BLS12-381

BLS12-381 [6], the current industry standard for ZK proof systems, achieves two-adicity 32 in both fields. Our construction achieves 36, enabling NTT of length $2^{36} \approx 6.8 \times 10^{10}$ —a $16\times$ improvement in maximum polynomial degree. The tradeoff is a larger ρ -value (2.0 vs. 1.5) and correspondingly larger field elements, which is acceptable for applications prioritizing long-term security (256-bit ECDLP vs. 128-bit for BLS12-381).

IV. READABILITY-AWARE PRIME SELECTION

A. Motivation

Cryptographic primes are typically opaque hexadecimal values that resist manual inspection. While the Cocks-Pinch method is inherently transparent—all parameters derive from a public seed T —the resulting primes still appear as random bit strings. This opacity complicates auditing: reviewers cannot easily verify structural properties or detect anomalies by inspection.

The Dual_EC_DRBG incident [7] demonstrated that even standardized constants can harbor backdoors when their generation process is opaque. We propose a complementary approach: among all cryptographically valid candidates, select the one that is most amenable to human inspection.

B. Security-First Design

Our readability scoring is a *post-filter* applied only to candidates that have already passed all security requirements:

1. Primality (Miller-Rabin with sufficient rounds)
2. Correct bit length (512 for r , 1024 for p)
3. Minimum two-adicity ($\nu_2(q-1) \geq 36$)
4. Valid CM equation ($4p = t^2 + 3y^2$)
5. Embedding degree verification ($r \mid \Phi_{18}(p)$)

The scoring system cannot weaken security because it only selects among pre-validated candidates.

TABLE II
READABILITY SCORING CRITERIA

Criterion	Rewards	Score Formula
Binary sparseness	Low Hamming weight	$(n - \text{hw}) \times 2$
Two-adicity	Higher $\nu_2(q-1)$	$\nu_2 \times 3$
Zero limbs	64-bit blocks = 0	count $\times 60$
Full limbs	Limbs = $0\text{xFF} \dots \text{F}$	count $\times 50$
Longest zero run	Contiguous zero limbs	run $\times 40$
Hex zero density	Hex digits = 0	$(z - 16) \times 2$
Simple top limb	Few bits in MSB limb	$(32 - \text{hw}) \times 3$
Repeating limbs	Adjacent equal limbs	pairs $\times 25$

TABLE III
READABILITY METRICS OF SELECTED PARAMETERS

Metric	p (1024-bit)	r (512-bit)
Hamming weight	464/1024	235/512
Two-adicity	36	36
Zero 64-bit limbs	0/16	8/16
Hex zero digits	33/256	16/128

C. Scoring Criteria

Each candidate prime is evaluated on eight independent structural criteria:

The combined score weights the base field prime p at $1.5\times$ the scalar field prime r , since p appears more frequently in implementations (field arithmetic, point encoding, serialization).

D. Search Procedure

The readability-aware search operates in two modes:

- **Fast mode:** Returns the first valid (p, r) pair (original behavior).
- **Readable mode:** For each valid r , explores the *full* lift range to find the highest-scoring p . Tracks the global best across N successful attempts (default $N = 50$) within a time budget.

In readable mode, the generator exhaustively searches all (h_t, h_y) combinations for each valid r , scoring every prime candidate that passes the security filter. The best-scoring (p, r) pair across all attempts is retained.

E. Results

The following table shows the readability metrics of our selected parameters. Both primes share the tail pattern $\dots \text{d970_00000001}$, a consequence of $T \equiv 0 \pmod{2^{12}}$ propagating through the construction. This shared structure aids visual verification.

The scalar field prime r achieves 8 zero limbs out of 16 (50%), meaning half of its 64-bit representation is zero—a property that simplifies implementation and testing. The shared tail $\text{d970_00000001} = d_{970} \cdot 2^{36} + 1$ visually confirms the NTT-friendly structure.

V. CURVE1024: PARAMETERS AND EVALUATION

This section presents the concrete Curve1024 parameters produced by our improved Cocks-Pinch construction with

TABLE IV
SCALAR FIELD PRIME r — LIMB DECOMPOSITION

Limb	Value	Note
15	0000000000000000	zero
14	0000000000000000	zero
13	0000000000000000	zero
12	0000000000000000	zero
11	0000000000000000	zero
10	0000000000000000	zero
9	0000000000000000	zero
8	0000000000000000	zero
7	c042c3389e72044d	
6	9ec8078aea7bc954	
5	ebf209d918f0ef27	
4	425259e247711927	
3	97463bf23832d0de	
2	5e46e47fc61cab50	
1	4bcfd36fc2a945c5	
0	1055d97000000001	tail: $d \cdot 2^{36} + 1$

TABLE V
BASE FIELD PRIME p — LIMB DECOMPOSITION

Limb	Value	Note
15	c0859da83a500988	
14	10eea35c61fea054	
13	c866902599220d30	
12	c93f38265ed6b830	
11	5785b8a073e482e8	
10	685cfeeb388dc45e	
9	bca1686245b326fe	
8	824ca74ac0844160	
7	f4e289806084e50e	
6	6c02f32cb94d786b	
5	f80fb080c07d6a07	
4	73f2b4591eb70df3	
3	9367aa32ebb42449	
2	e0a648f61a20e876	
1	f971f3bf976a7d6a	
0	dd20d97000000001	tail: $d \cdot 2^{36} + 1$

readability-aware selection, followed by implementation details and security analysis.

A. Curve1024 Parameters

The construction produces the following pairing-friendly elliptic curve over a 1024-bit prime field. All parameters are publicly verifiable from the seed T .

1) *Seed and Construction*: The curve is fully determined by a single 86-bit seed:

$$T = 0x26704d2ace0facdd539 \cdot 2^{12}$$

This seed satisfies $T \equiv 0 \pmod{2^{12}}$, guaranteeing $\nu_2(r - 1) = 36$ via the factorization $r - 1 = T^3(T^3 - 1)$.

2) *Scalar Field Order (512 bits)*: The scalar field prime $r = \Phi_{18}(T) = T^6 - T^3 + 1$ decomposes as:

$$r = d_r \cdot 2^{36} + 1$$

where d_r is a 476-bit odd integer. In 64-bit limb representation (MSB first):

Key readability features: 8 of 16 limbs are zero (50%), and the final limb visually confirms the NTT-friendly structure with the 00000001 suffix (36 trailing zero bits before the +1).

3) *Base Field Prime (1024 bits)*: The base field prime $p = (t^2 + 3y^2)/4$ decomposes as:

$$p = d_p \cdot 2^{36} + 1$$

where d_p is a 988-bit odd integer. The final limb shares the same tail pattern as r :

Shared tail pattern: Both p and r end with ...d970_00000001, a direct consequence of the seed constraint propagating through the construction. This visual consistency aids manual verification.

TABLE VI
CURVE1024 PARAMETERS

Parameter	Value
Equation	$E : y^2 = x^3 + 41$
Base field $ p $	1024 bits
Scalar field $ r $	512 bits
Embedding degree k	18
CM discriminant D	-3 (hence $j = 0$)
$\rho = \log p / \log r$	2.0
$\nu_2(p - 1)$	36
$\nu_2(r - 1)$	36
Cofactor $h = \#E/r$	$\approx 2^{512}$

4) *Curve Summary*: The curve equation $y^2 = x^3 + 41$ arises from the CM method with $D = -3$: the Hilbert class polynomial $H_{-3}(x) = x$ gives $j = 0$, yielding the family $y^2 = x^3 + b$. The twist $b = 41$ is selected by verifying which of the six sextic twists has group order divisible by r .

A generator point $G = (G_x, G_y)$ of prime order r is obtained by cofactor multiplication of a random curve point. The full coordinates are published in the project repository [9].

5) *Readability Metrics*: The scalar field prime r is notably sparse: half its limb representation is zero, making it easy to recognize and verify in debugging output. Both primes exceed BLS12-381's two-adicity of 32.

B. Implementation

We implement the complete system in Rust without external cryptographic dependencies [9]. The implementation comprises:

- A custom U1024 big integer type with Montgomery multiplication [10]
- Prime field arithmetic (PrimeFieldElement) with constant-time Fermat inversion
- Elliptic curve group operations in affine coordinates
- Schnorr and ECDSA signature schemes
- The parameter generation tool (improved Cocks-Pinch + readability scoring)

TABLE VII
READABILITY COMPARISON

Metric	p (1024-bit)	r (512-bit)
Hamming weight	464/1024 (45%)	235/512 (46%)
Two-adicity	36	36
Zero 64-bit limbs	0/16	8/16 (50%)
Hex zero digits	33/256 (13%)	16/128 (12.5%)
Trailing zero bits	36	36

TABLE VIII
PERFORMANCE BENCHMARKS (CRITERION, RELEASE BUILD, LLVM O3,
NO AVX-512)

Operation	Min	Mean	Max
Key generation	1.629 s	1.687 s	1.757 s
Schnorr sign	1.473 s	1.551 s	1.641 s
Schnorr verify	2.656 s	2.831 s	3.025 s
ECDSA sign	1.524 s	1.606 s	1.711 s
ECDSA verify	2.847 s	3.140 s	3.459 s
Scalar mult. $[k]G$	1.061 s	~1.2 s	1.489 s

The test suite contains 36 tests covering arithmetic correctness, group law verification, and attack simulation (MOV, Anomalous, Invalid Curve).

C. Performance Benchmarks

The performance reflects the 1024-bit field size: each scalar multiplication requires ~512 double-and-add iterations, each involving a Fermat inversion (~1536 Montgomery multiplications). This is approximately $8000\times$ slower than optimized secp256k1 implementations, which is expected given the $4\times$ larger field and $O(n^2)$ multiplication cost. The system targets offline signing scenarios (legal documents, software releases, PKI certificates) where seconds of latency are acceptable.

D. Security Analysis

The 256-bit ECDLP security exceeds NIST SP 800-57 recommendations through 2030+ [11]. The MOV attack [12] maps ECDLP to DLP in $\mathbb{F}_{p^{18}}$, a field of 18,432 bits—far exceeding the 3,072-bit threshold for 128-bit security. The curve is provably non-anomalous [13] since $\#E(\mathbb{F}_p) = h \cdot r$ with $r \approx 2^{512} \neq p \approx 2^{1024}$.

Limitation: The current affine-coordinate implementation uses variable-time double-and-add, which leaks scalar bits through timing. A Montgomery ladder (constant-time) is identified as future work.

E. Curve Family Comparison

Curve1024 occupies a unique position: it provides the highest ECDLP security level (256-bit) and the highest two-adicity (36) among deployed pairing-friendly curves, at the cost of a larger ρ -value inherent to the Cocks-Pinch method. The 18,432-bit pairing target field provides substantial margin against TNFS attacks.

TABLE IX
SECURITY EVALUATION SUMMARY

Attack	Complexity	Status
Pollard- ρ (ECDLP)	$O(2^{256})$	Secure
MOV reduction	DLP in $\mathbb{F}_{p^{18}}$ (18432-bit)	Secure
Anomalous (SSSA)	Requires $\#E = p$	Immune
TNFS [BarbulescuDuquesne2019]	≥ 250 -bit in $\mathbb{F}_{p^{18}}$	Secure
Invalid curve	Point validation in constructor	Mitigated

TABLE X
COMPARISON WITH STANDARD PAIRING-FRIENDLY CURVE FAMILIES

Curve	$ p $ (bits)	ρ	ν_2	ECDLP Security	Pairing Target
BN254	254	1.0	~1	100-bit	3048-bit
BLS12-381	381	1.5	32	128-bit	4572-bit
KSS18 (generic)	varies	1.33	random	varies	varies
Curve1024	1024	2.0	36	256-bit	18432-bit

VI. RELATED WORK

Pairing-friendly curve taxonomy. Freeman, Scott, and Teske [5] provide a comprehensive classification of pairing-friendly curve constructions, including the Cocks-Pinch method, MNT curves, and polynomial families (BN, BLS, KSS). Their work establishes the theoretical framework but does not address NTT-friendliness as a design criterion.

TNFS security updates. Barbulescu and Duquesne [14] demonstrated that the Tower Number Field Sieve reduces the effective security of many pairing-friendly curves, notably downgrading BN256 from 128-bit to approximately 100-bit security. This motivates our choice of a 1024-bit base field with $k = 18$, yielding a pairing target of 18,432 bits.

BLS12-381. Bowe [6] designed BLS12-381 specifically for zk-SNARKs, achieving two-adicity 32 in both fields through careful parameter selection within the BLS12 family. Our work achieves higher two-adicity (36) using the more flexible Cocks-Pinch method, at the cost of a larger ρ -value.

Pasta curves. The Pallas/Vesta cycle [15] achieves NTT-friendliness through a 2-cycle of curves, enabling recursive proof composition. Unlike our approach, Pasta curves are not pairing-friendly and target a different application domain (Halo 2 recursive proofs).

Nothing-up-my-sleeve numbers. The cryptographic community has long advocated for “nothing-up-my-sleeve” constants (e.g., using digits of π) to demonstrate absence of backdoors. Our readability scoring takes a complementary approach: rather than deriving primes from a fixed public constant, we select the most inspectable prime from a large space of equally secure candidates. Both approaches serve the goal of transparent parameter generation.

Cocks-Pinch extensions. Chen, Hong, and Wang [16] extend the Cocks-Pinch method to embedding degrees 5–8 with optimal Ate pairing computation. Our work focuses on $k = 18$ with the novel addition of NTT constraints and readability scoring.

VII. CONCLUSION

We presented three contributions to pairing-friendly elliptic curve construction:

1. **NTT-friendly Cocks-Pinch.** By constraining the cyclotomic seed $T \equiv 0 \pmod{2^{12}}$, we guarantee two-adicity ≥ 36 in the scalar field deterministically, while an extended lift search achieves the same for the base field. This exceeds the industry-standard BLS12-381 (two-adicity 32) at a modest increase in generation time (30–60 s vs. 2–10 s).
2. **Readability-aware prime selection.** Our multi-criteria scoring system produces primes that are more amenable to human inspection without any security degradation, addressing the auditability concern in cryptographic parameter generation.
3. **Curve1024.** We instantiate these techniques into a concrete, publicly verifiable curve: $E : y^2 = x^3 + 41$ over a 1024-bit prime field with embedding degree 18, providing 256-bit ECDLP security and NTT support up to length 2^{36} in both fields. The complete parameter set (field primes, generator point, CM data) is published and validated by a Rust implementation with 36 passing tests.

The system is suitable for offline signing applications requiring long-term security guarantees and for future pairing-based protocols (BLS aggregate signatures, zk-SNARKs) that benefit from the NTT-friendly field structure.

Future work includes: (1) projective/Jacobian coordinates to eliminate per-operation inversions, (2) constant-time Montgomery ladder for side-channel resistance, (3) full optimal Ate pairing implementation over $\mathbb{F}_{p^{18}}$, and (4) extension to other embedding degrees ($k = 12, 24$).

A. Availability

The implementation and all curve parameters are available as open-source software under the MIT license [9].

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